August
2026

PASSWORD CRACKING WITH JTR

AND

NETWORKWALKS TOOLS

CYBERSECURITY & ETHICAL HACKING PROJECT TASKS

B082-Networkwalks

WEEK 3

PROJECT MODULES: 1 & 2

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1. Introduction

This report covers two modules, the first module covers password cracking with John the Ripper (JTR) (W3-PM1), while the second module covers password cracking with Networkwalks tools (W3-PM2).

Many files like PDF, ZIP, and Office documents can be locked with a password. When a file is locked, its password is stored in the form of a hash. A hash is a scrambled value that represents the password. To recover the password, we first take out this hash from the file and then run it through a cracking tool that tries different words until it finds a match.

In this lab tasks (W3-PM1) I used JTR John and JTR Johnny to recover the passwords of the protected PDF files. I also used two free online tools made by Networkwalks, namely, the Hash Calculator and Password Cracker. First used the Hash Calculator to take the hash out of the locked PDF files. Then i used the Password Cracker to find the real password from the hash.

2. Definitions and background

Password cracking is the process of recovering a password from stored data or a protected file. Security professionals use it to test how strong a password is and to show why weak passwords are risky.

John the Ripper (JTR) is a popular password cracking tool used by security professionals to test how strong passwords are. It started as a tool for Unix systems but now works on Windows, Linux, and Mac. It can check many types of password hashes and also unlock password protected files like PDF, ZIP, and Office documents.

Johnny is the graphical version of John the Ripper. It gives a simple point and click screen, so beginners can use JTR without typing long commands. Both tools are widely used in security testing and learning labs to understand password safety.

3. Modules completed & Tools utilized

The modules completed and tools utilized are listed in the table below:

Modules	**Tools/resources**
W3-PM1	John the Ripper (JTR) (Terminal) & Johnny (GUI) JTR_default_password.txt, Online HashCrack, Notepad, and Windows 11 Laptop)
W3-PM2	Online Networkwalks Hash Calculator & Password Cracker, Notepad, and Windows 11 Laptop

4. Targeted Files

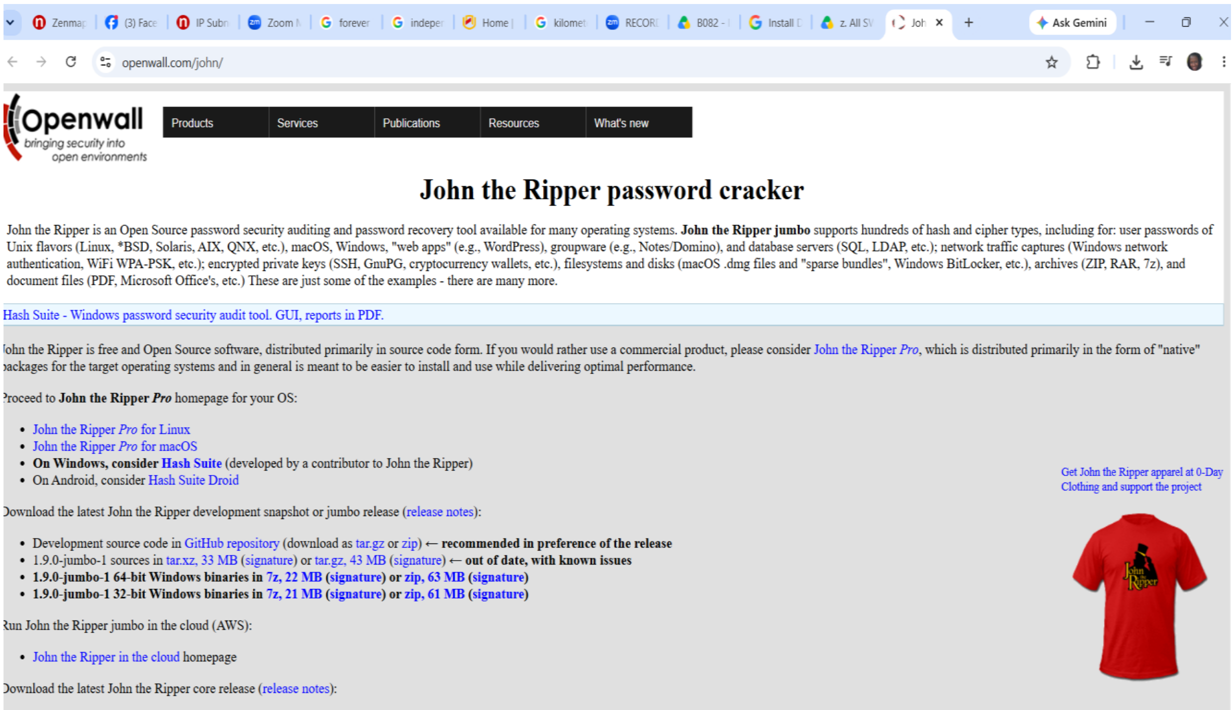
Modules	**Password cracked**	**Status**
My Locked PDF1.pdf	**good-luck**	Password successfully cracked ☑
My Locked PDF2.pdf	**password1**	Password successfully cracked ☑
My Locked PDF3.pdf	**1qaz2wsx**	Password successfully cracked ☑

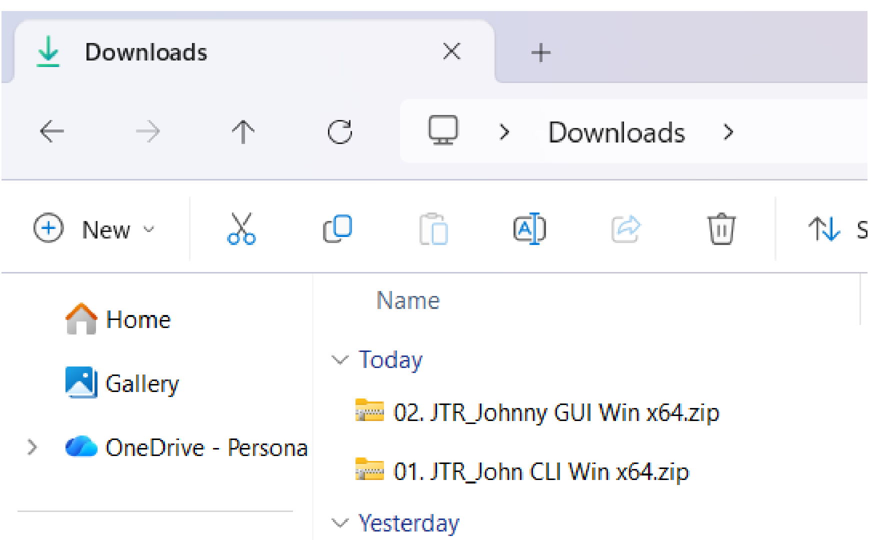
5. Activities Performed

5.1 Module 1: Password cracking with John the Ripper

STEP 1

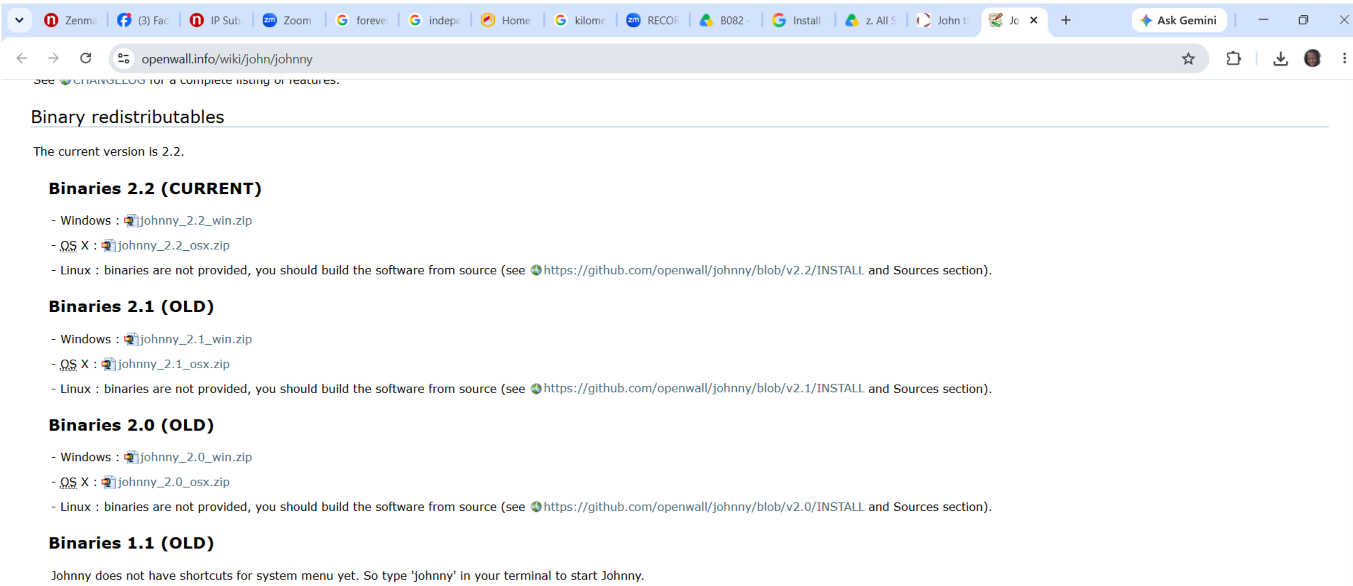
I downloaded John the Ripper from the official website on my windows PC as per the below screenshots.



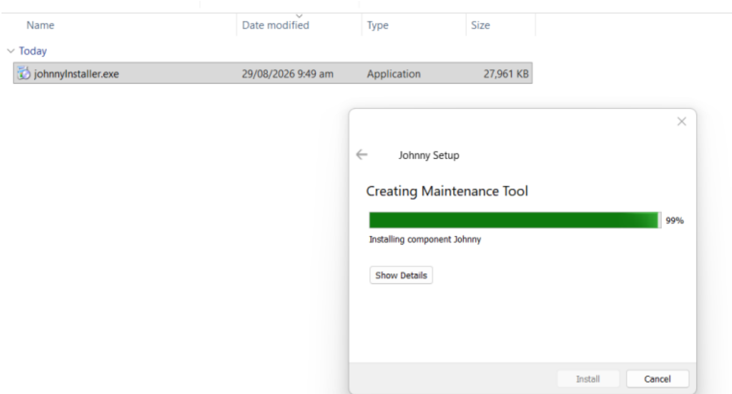


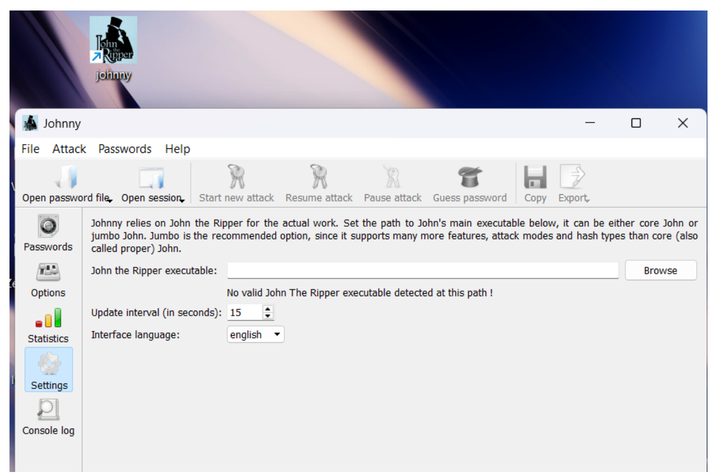
STEP 2

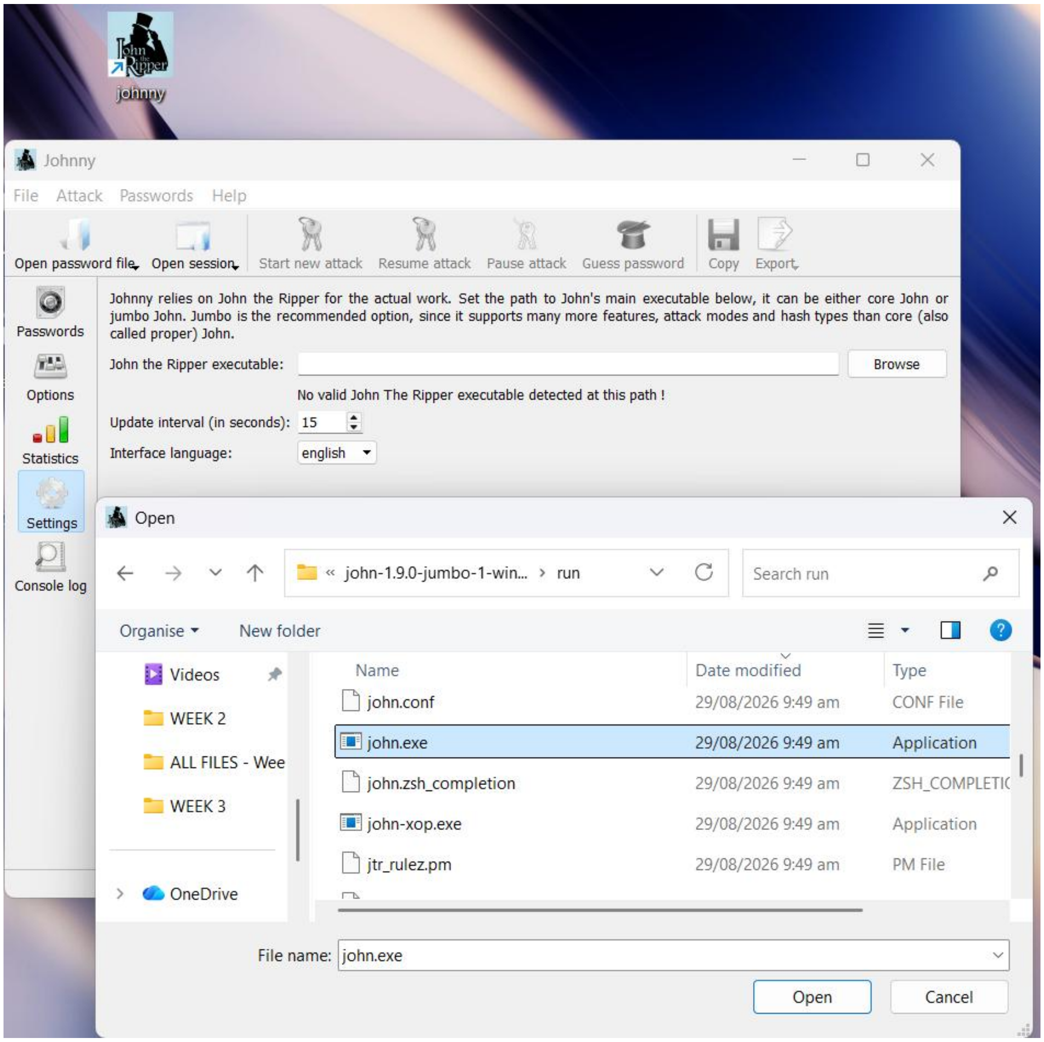
I downloaded and installed Johnny GUI from the official website as shown below:



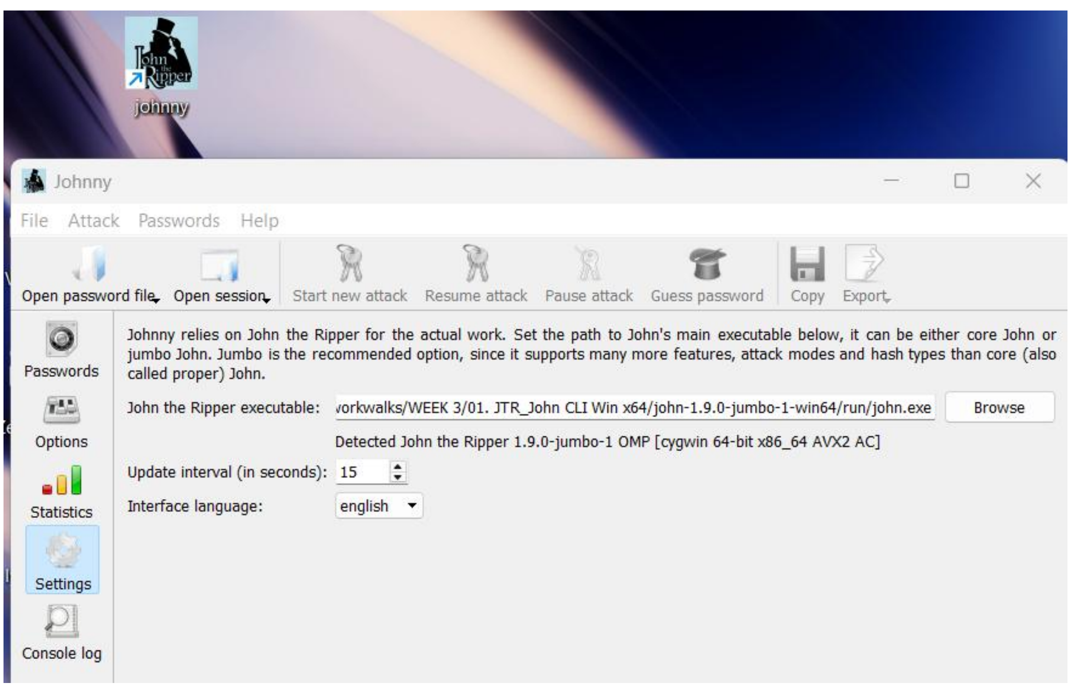
Run the setup file & install Johnny as shown in below:







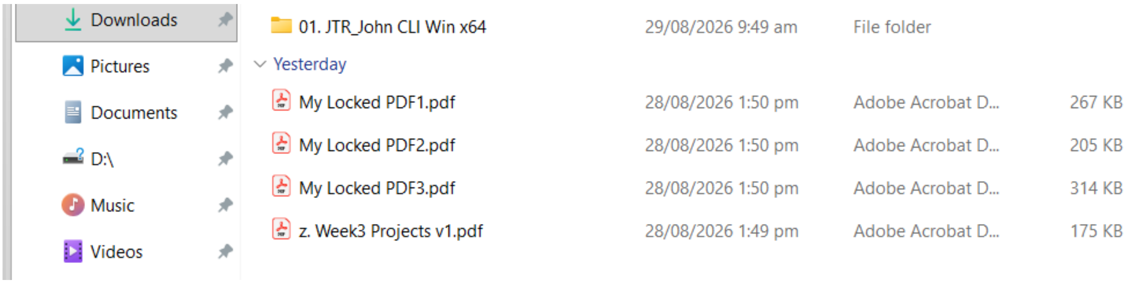
Select John.exe: **john.exe file is located in the run folder as shown in this screenshot.*



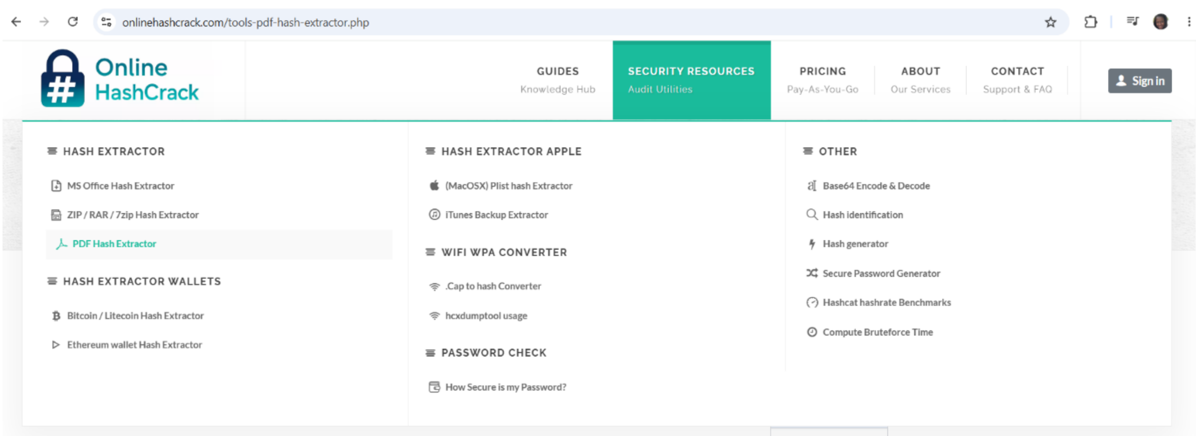
STEP 3

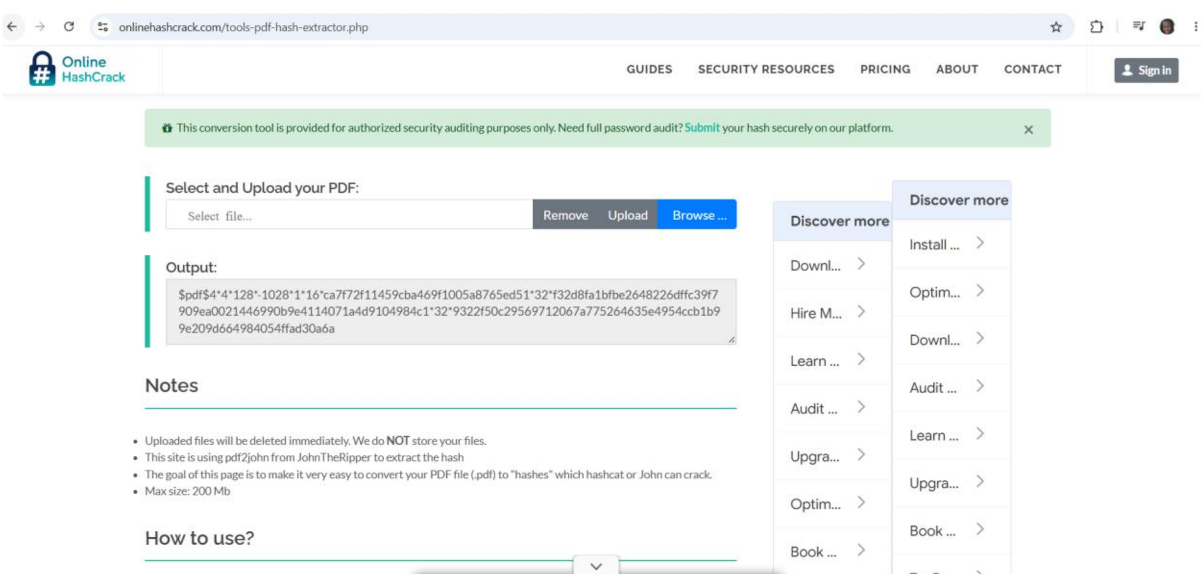
I follow the below steps to crack the password.

I downloaded the encrypted PDF files to my laptop/PC:



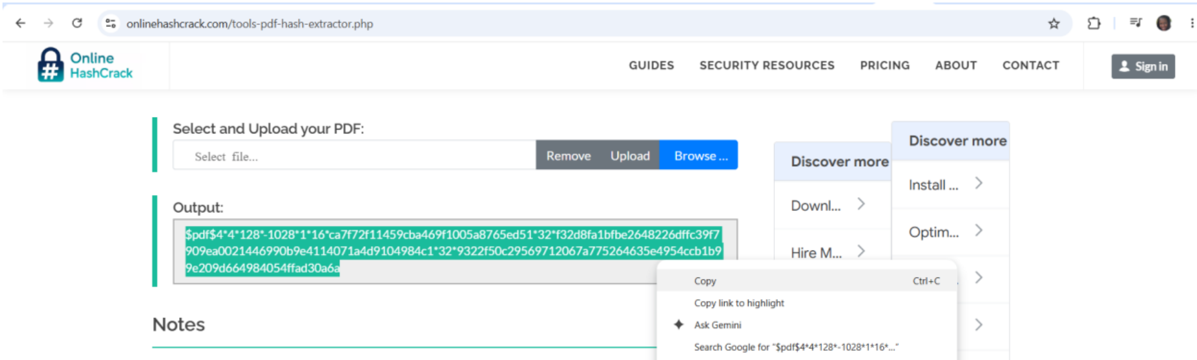
I then opened the hash website and uploaded the pdf files to find the hash:
https://www.onlinehashcrack.com/tools-pdf-hash-extractor.php



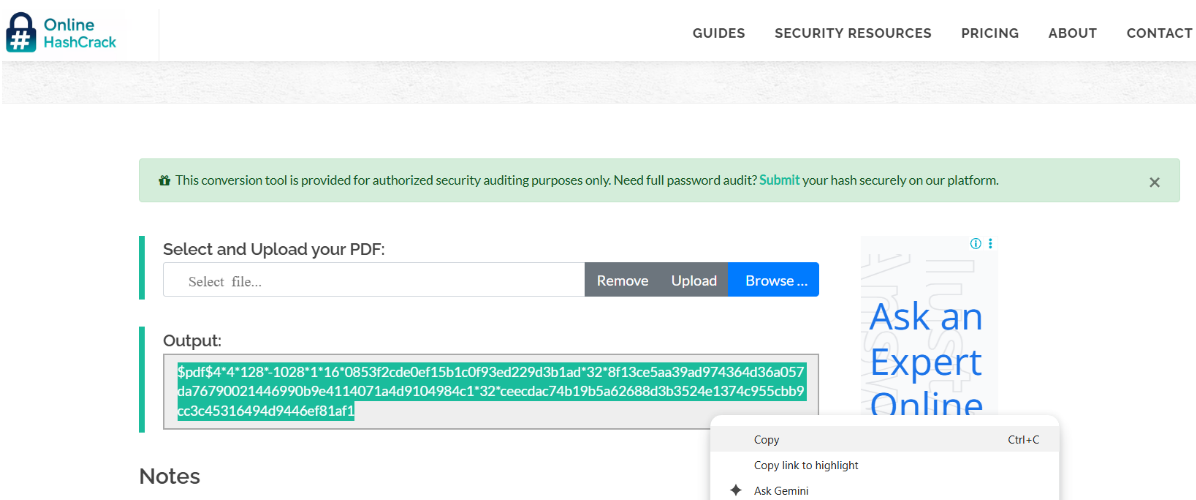


Browse the PDF file and clicked on Upload:

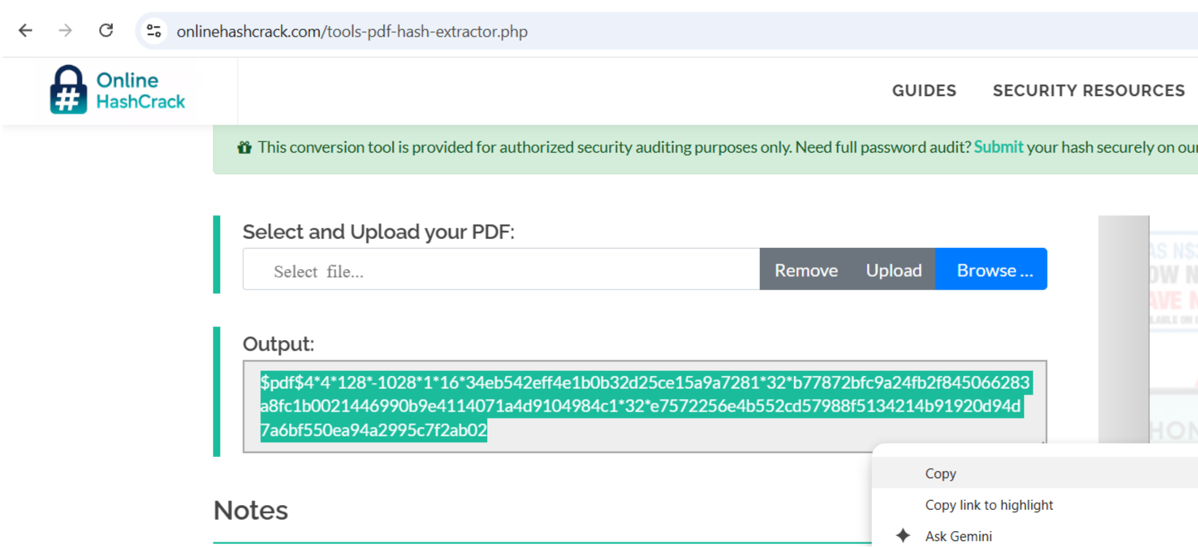
I copied the hash values and thereafter saved them in Note Pad as indicated below:



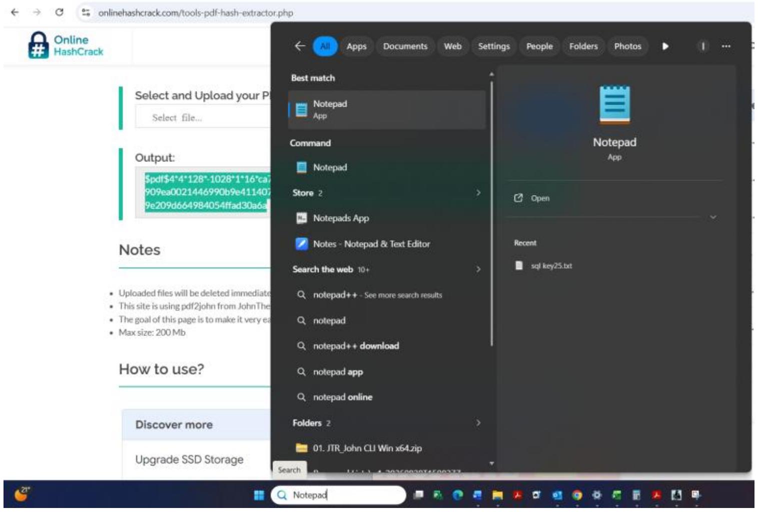
My Locked PDF1 hash value



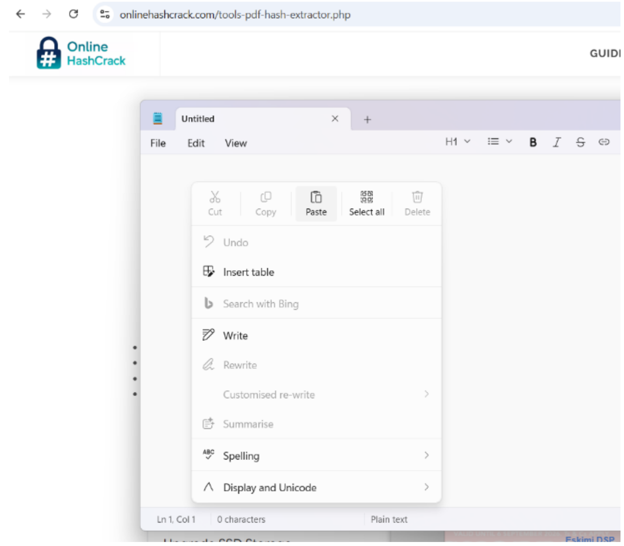
My Locked PDF2 hash value



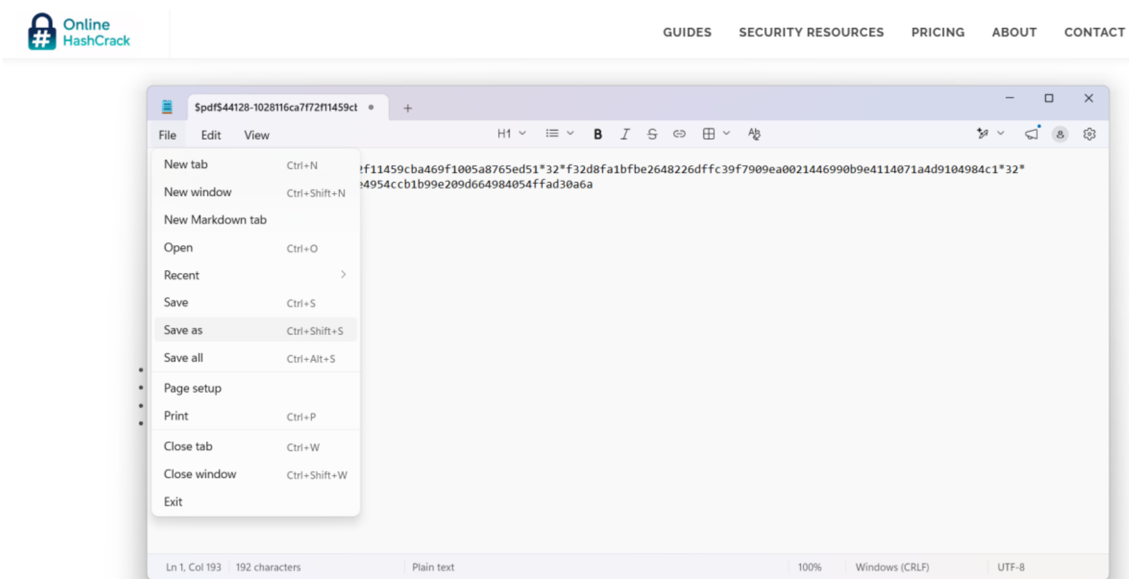
My Locked PDF3 hash value



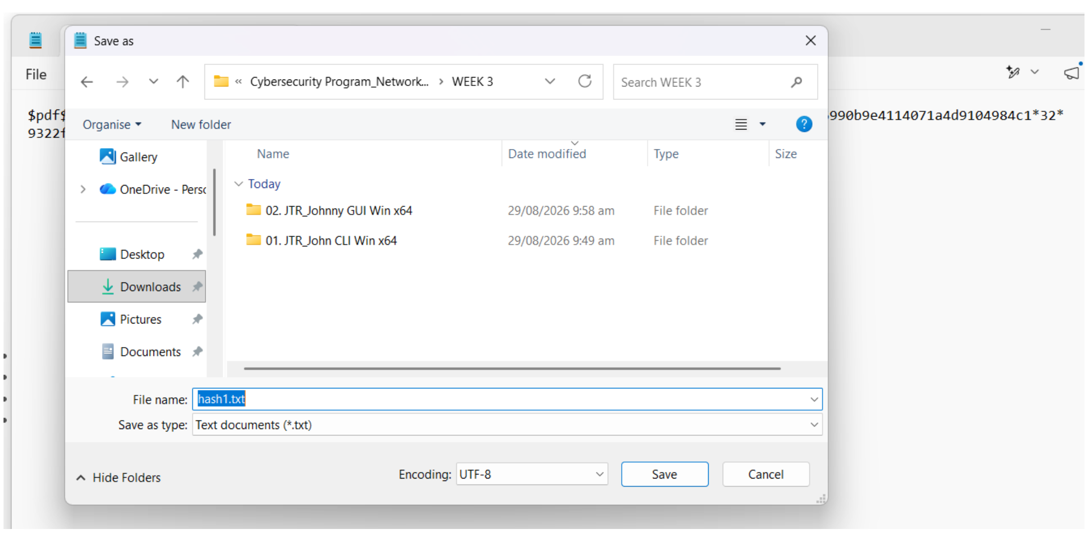
Opening notepad



Pasting the hash values

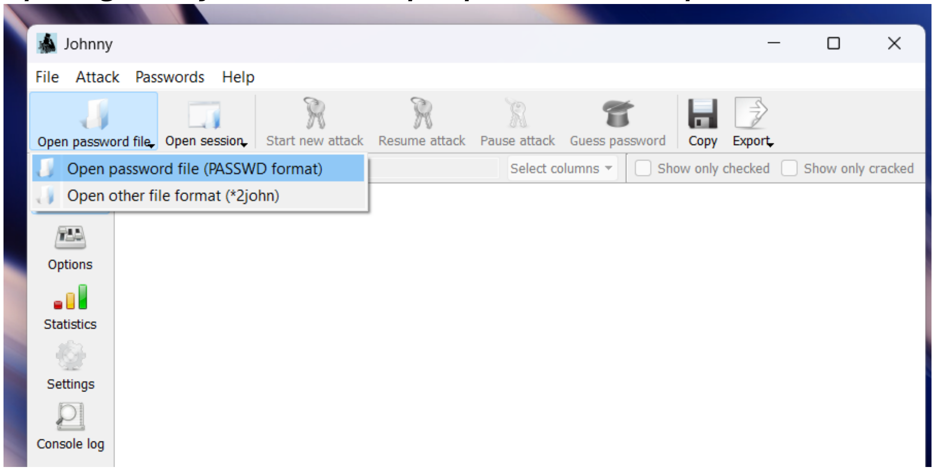


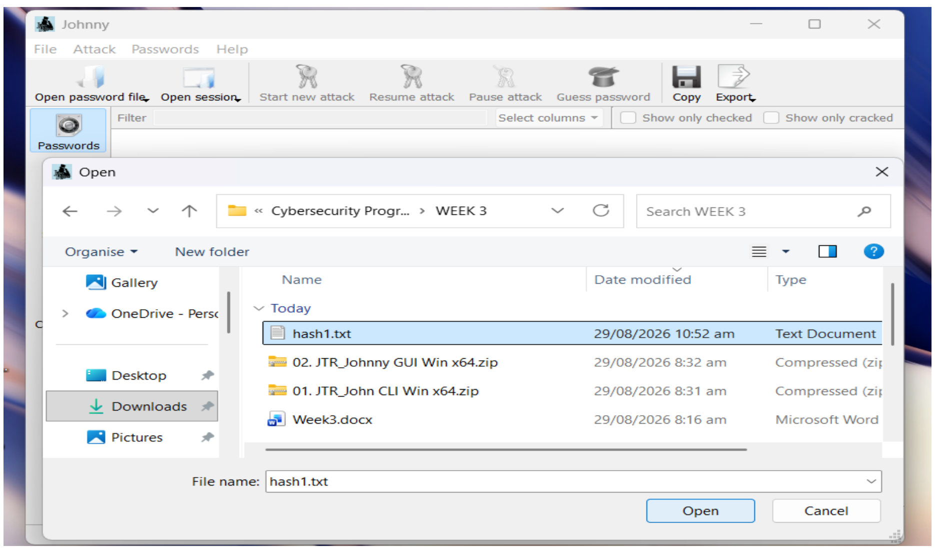
Saving the text file



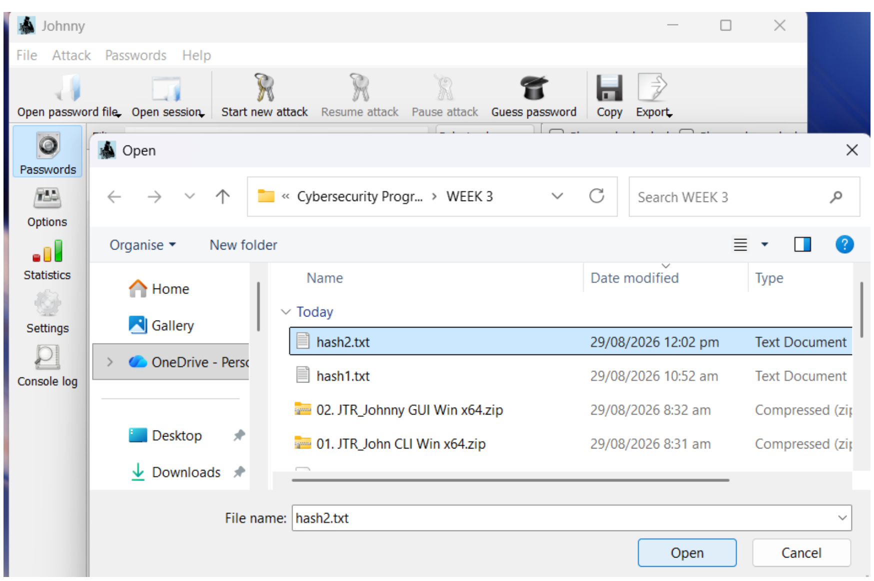
Saving the file with name as *hash1.txt*

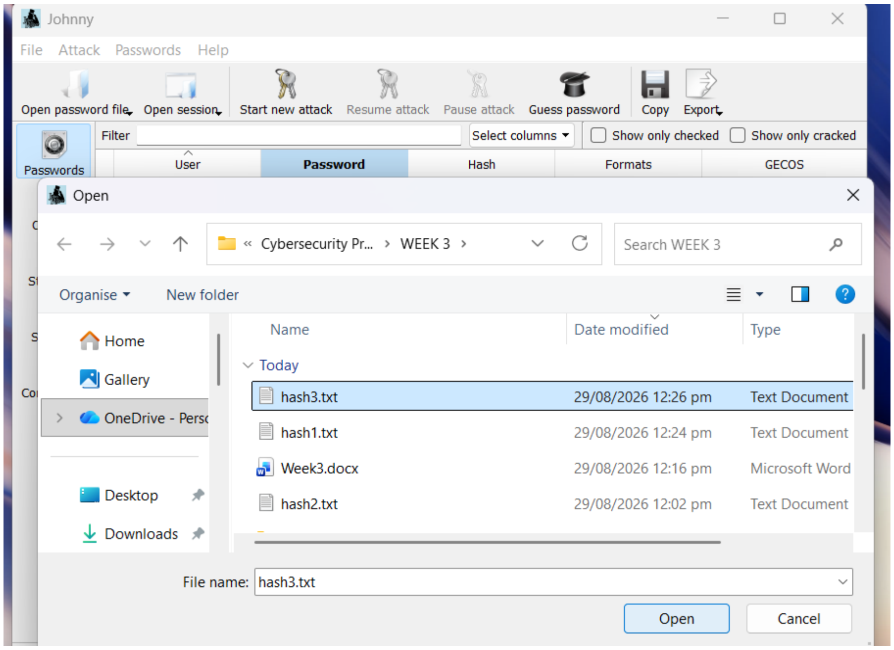
Opening Johnny : i clicked on open password file as per the screenshot below.

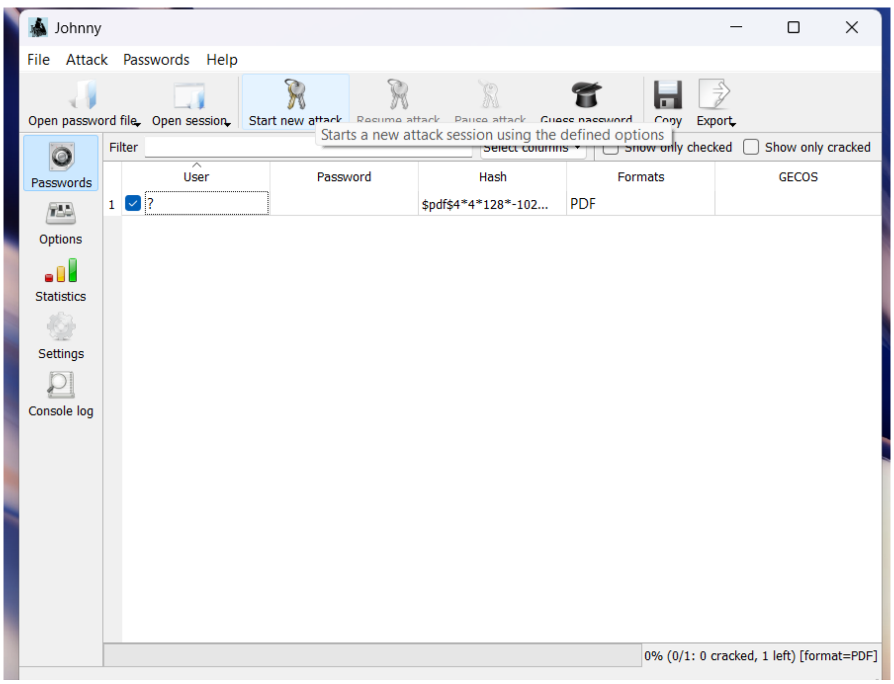




Browsed to the hash .txt file and clicked on open

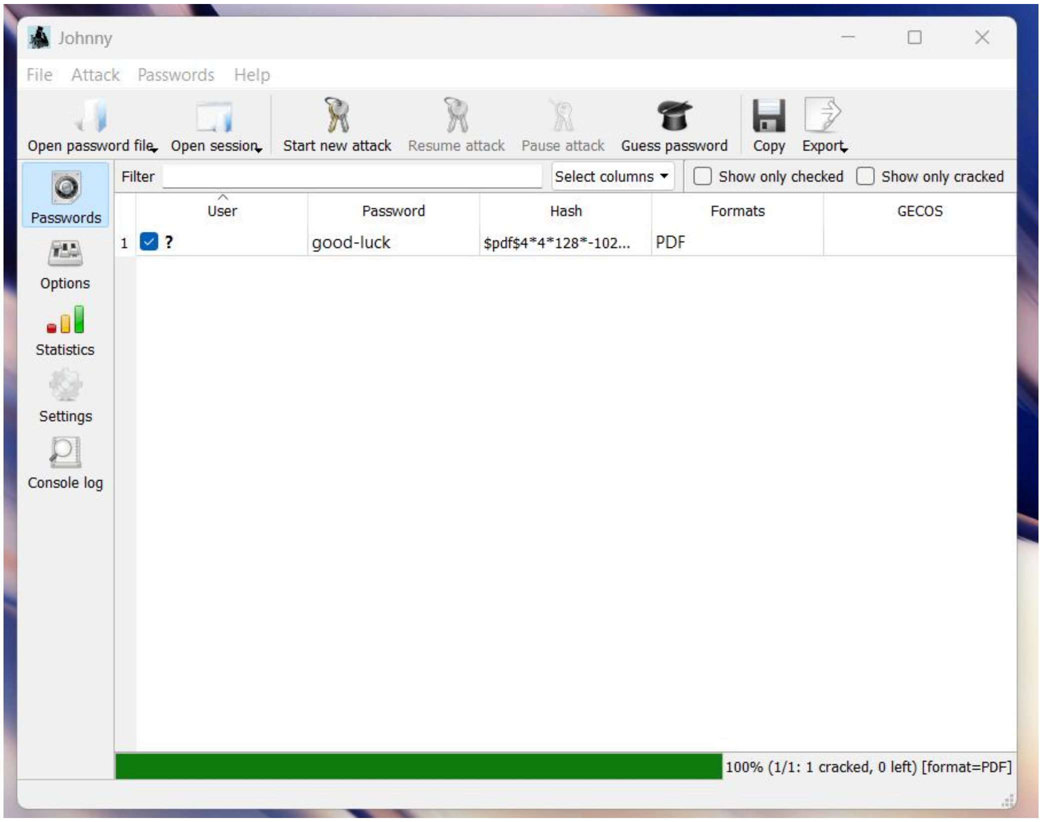




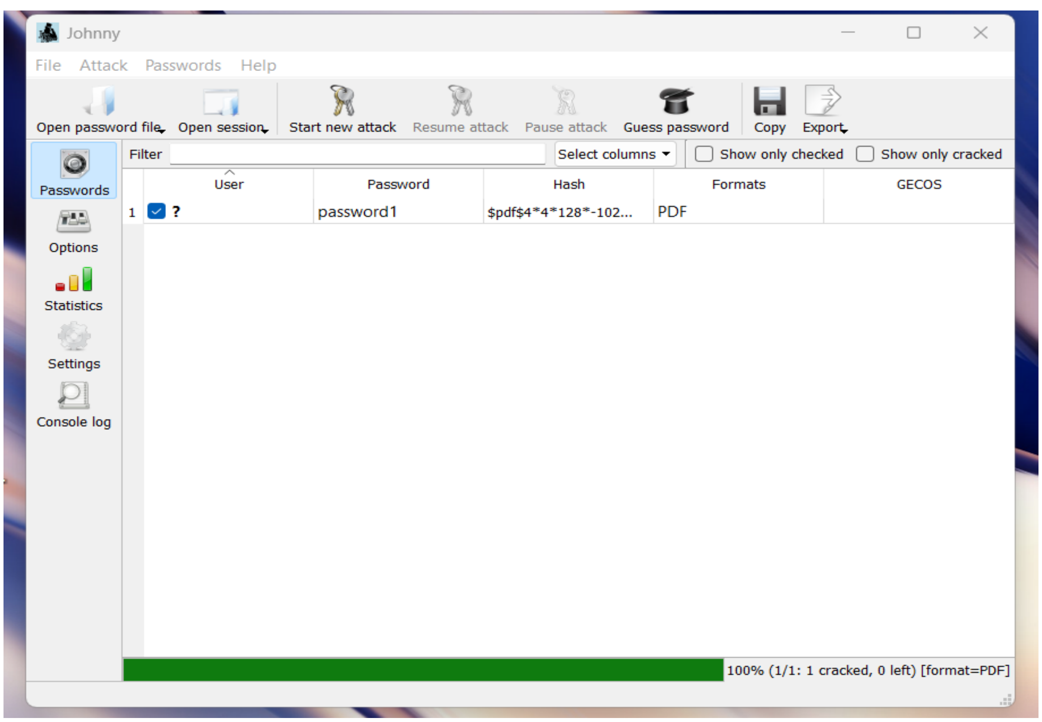


Clicked on '*Start new attack*'

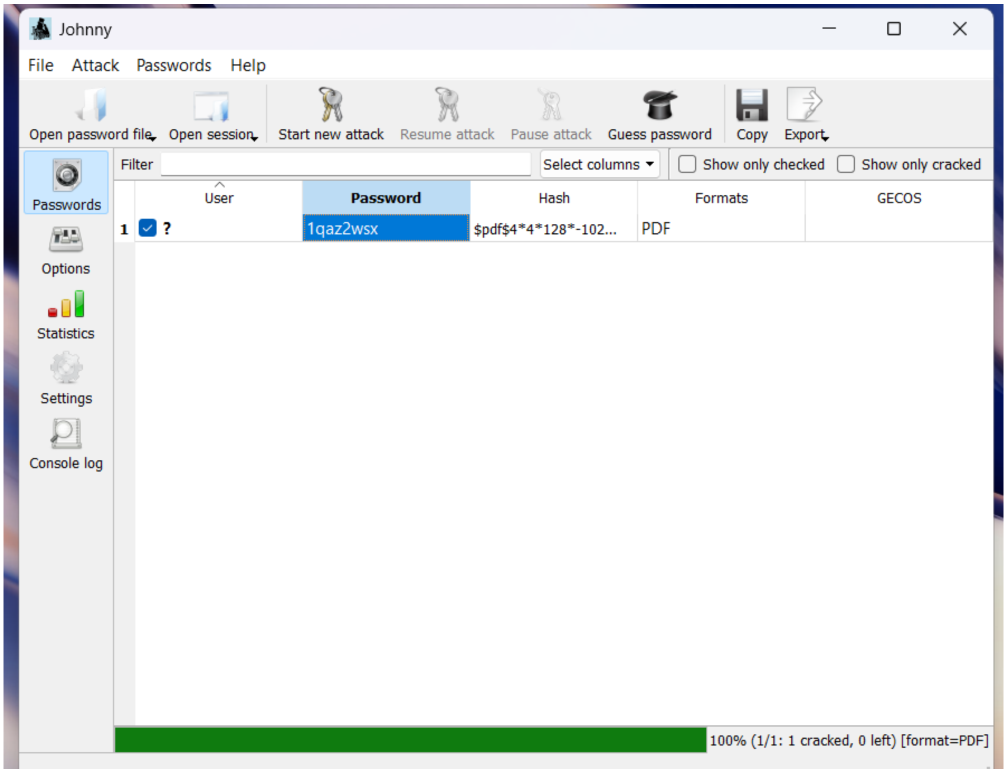
The passwords were cracked and used to open the locked PDF files as per the below screen shots.



My Locked PDF1 password cracked as: good-luck

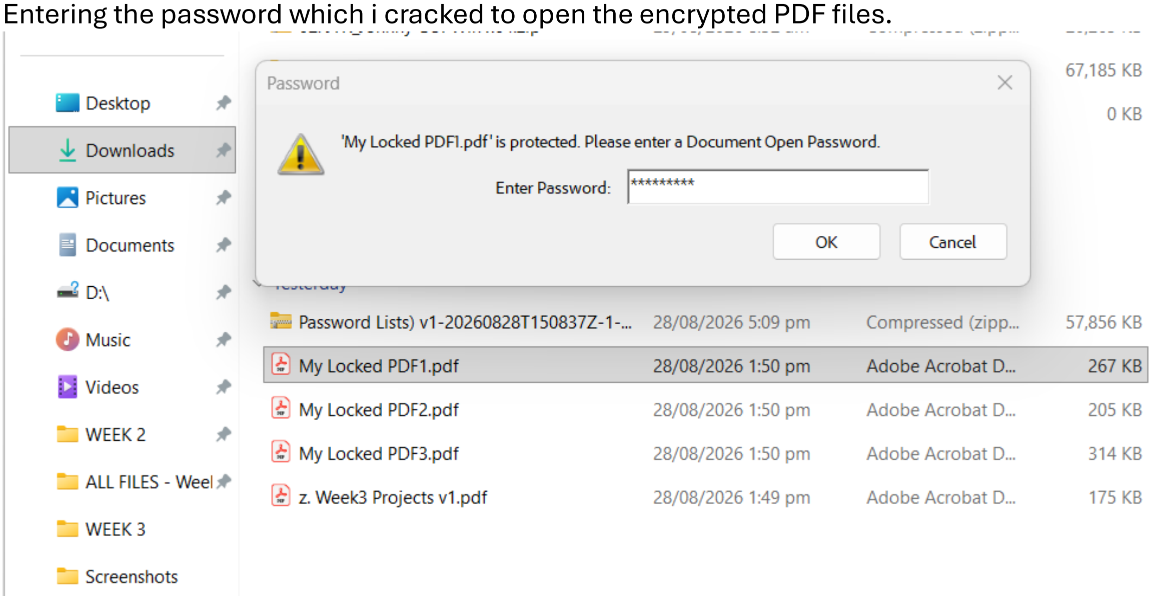


My Locked PDF2 password cracked as: password1

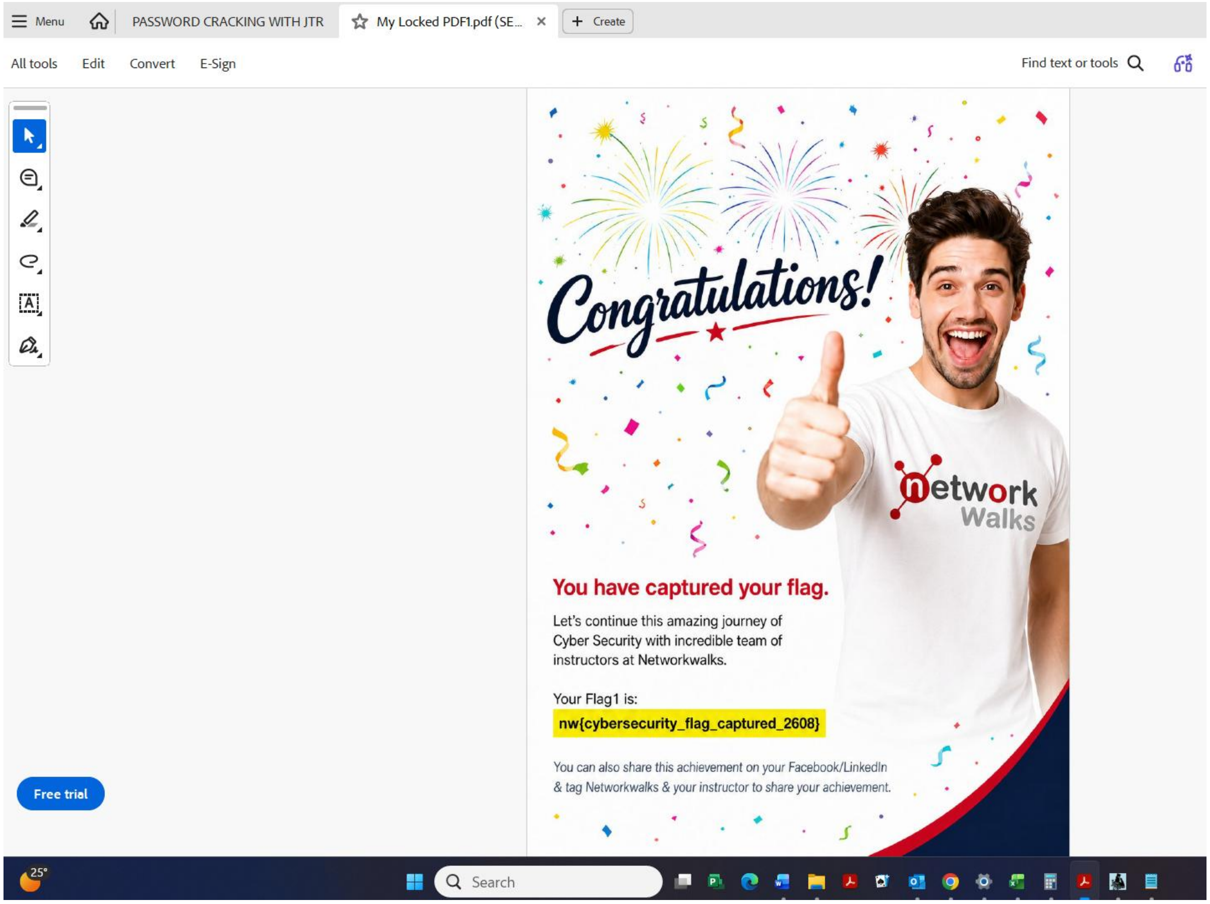


My Locked PDF3 password cracked as: 1qaz2wsx

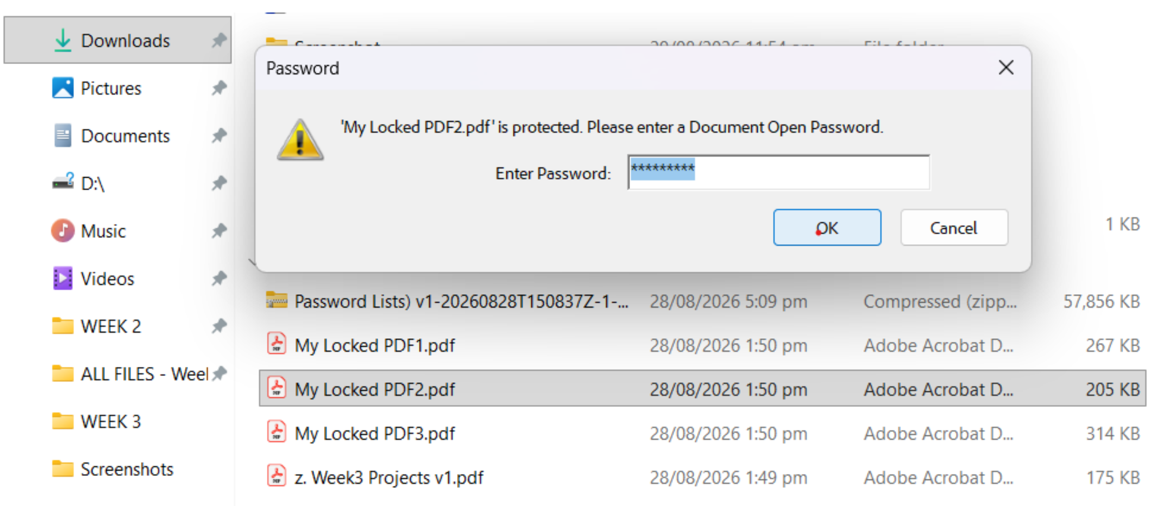
Entering the password which i cracked to open the encrypted PDF files.



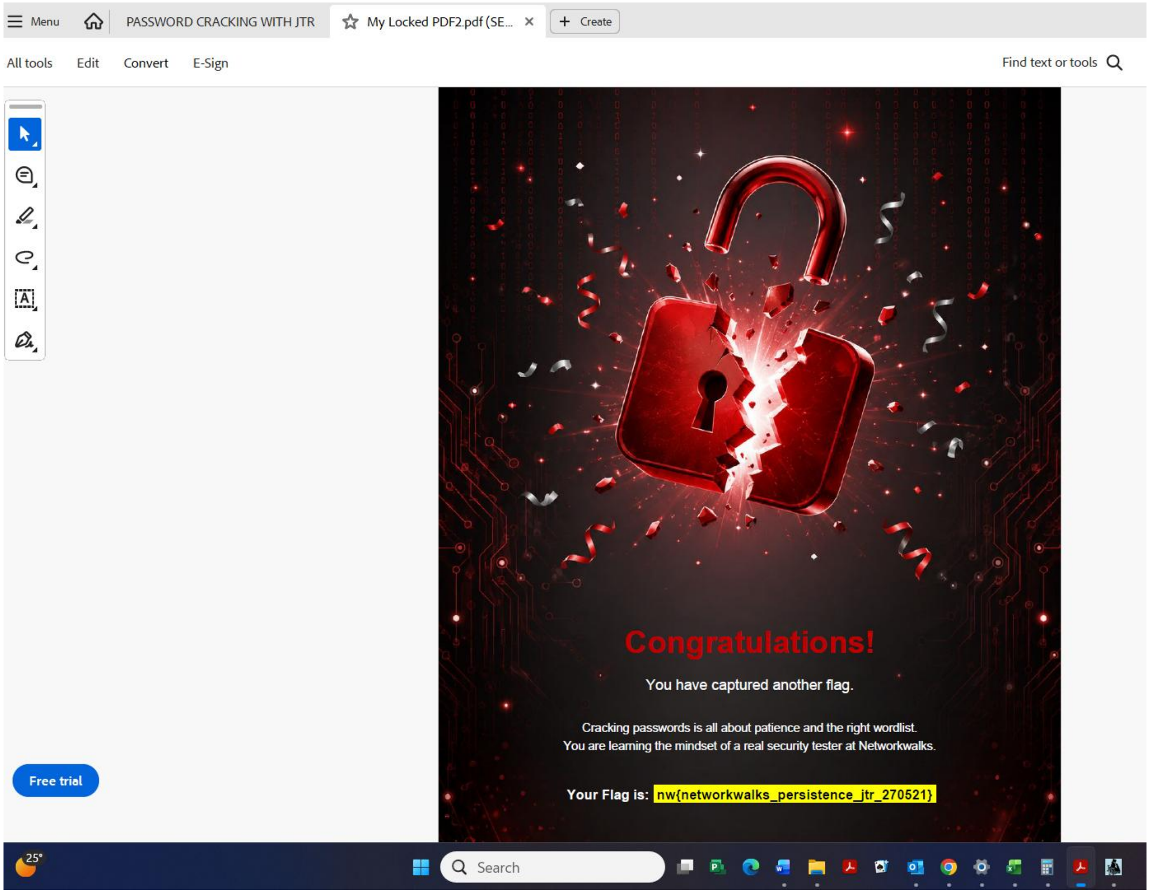
Opening My Locked PDF1



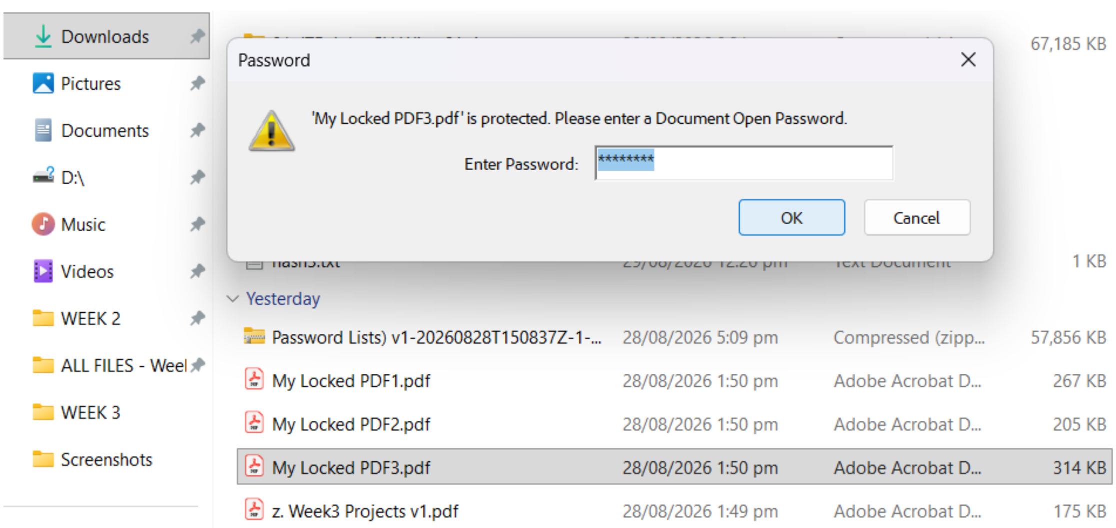
Opened "My Locked PDF1"



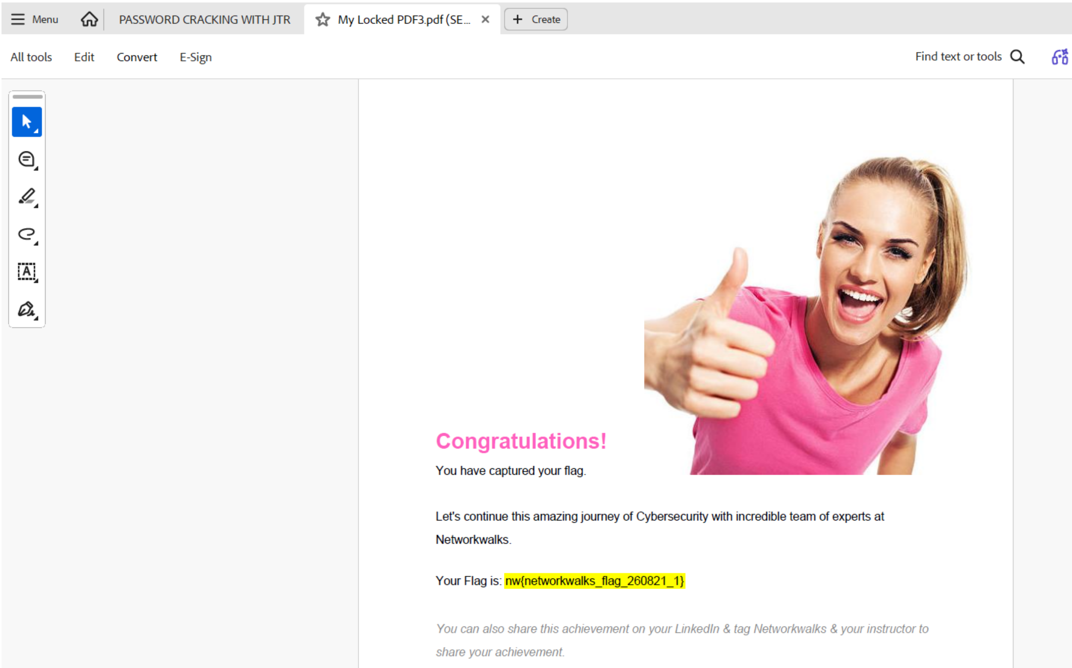
Opening My Locked PDF2



Opened "My Locked PDF2"



Opening My Locked PDF3



Opened "My Locked PDF3"

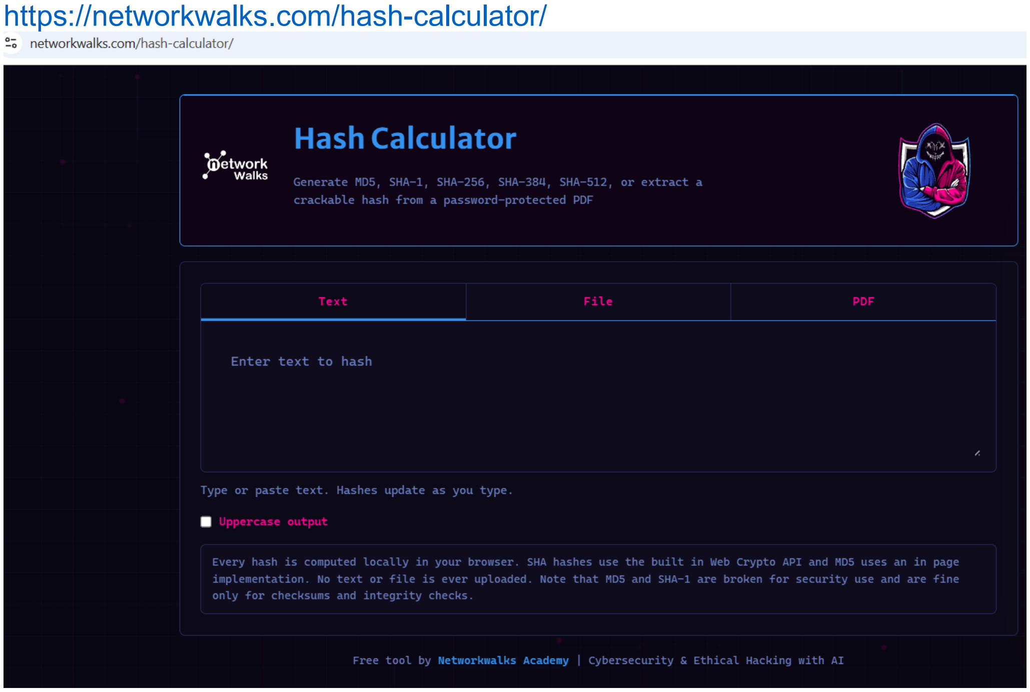
5.2 Module 2: *Password cracking with Networkwalks Tools*

STEP 1

I Download the encrypted PDF files to my laptop from the lab page using the below link:
https://networkwalks.com/project-task-lab-password-cracking-with-networkwalks-tools/

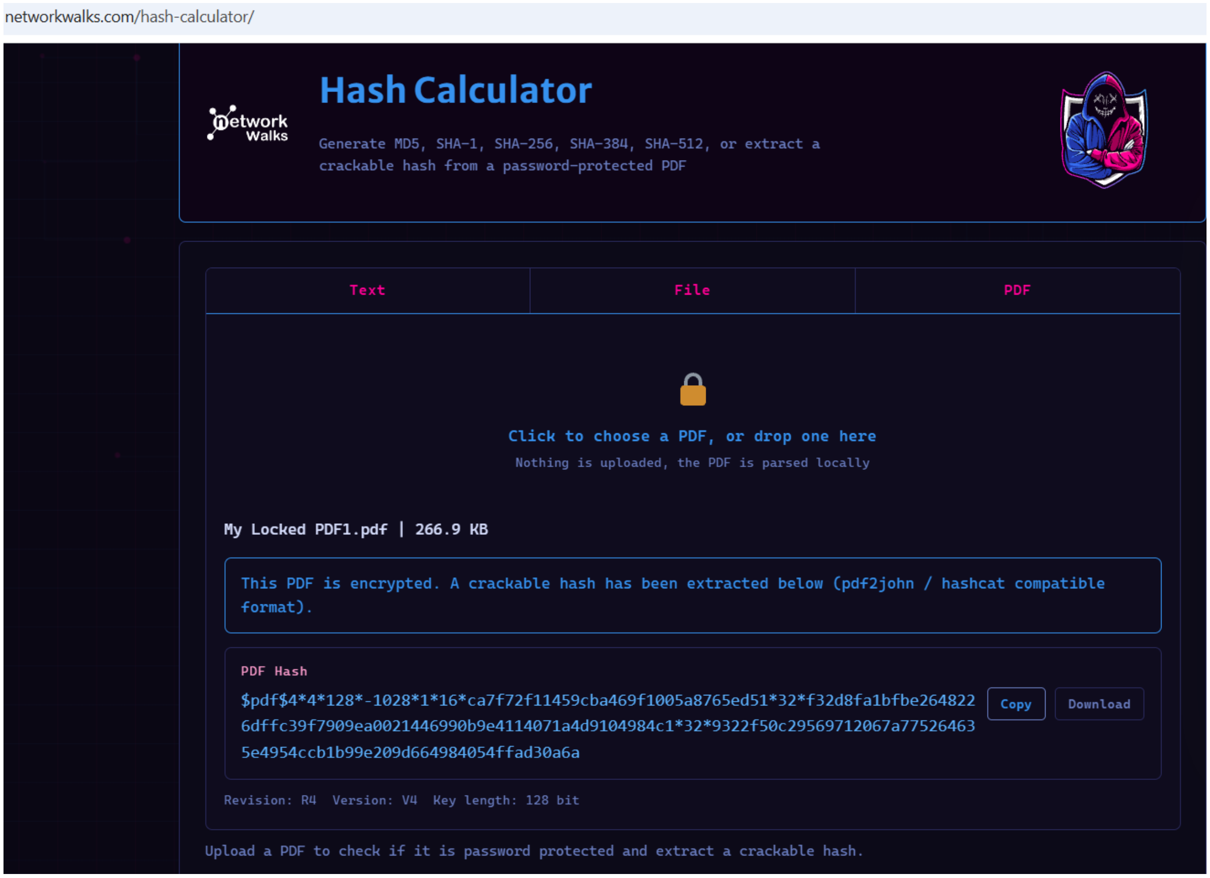
STEP 2

I opened the Networkwalks Hash Calculator in the web browser as per the below:
https://networkwalks.com/hash-calculator/



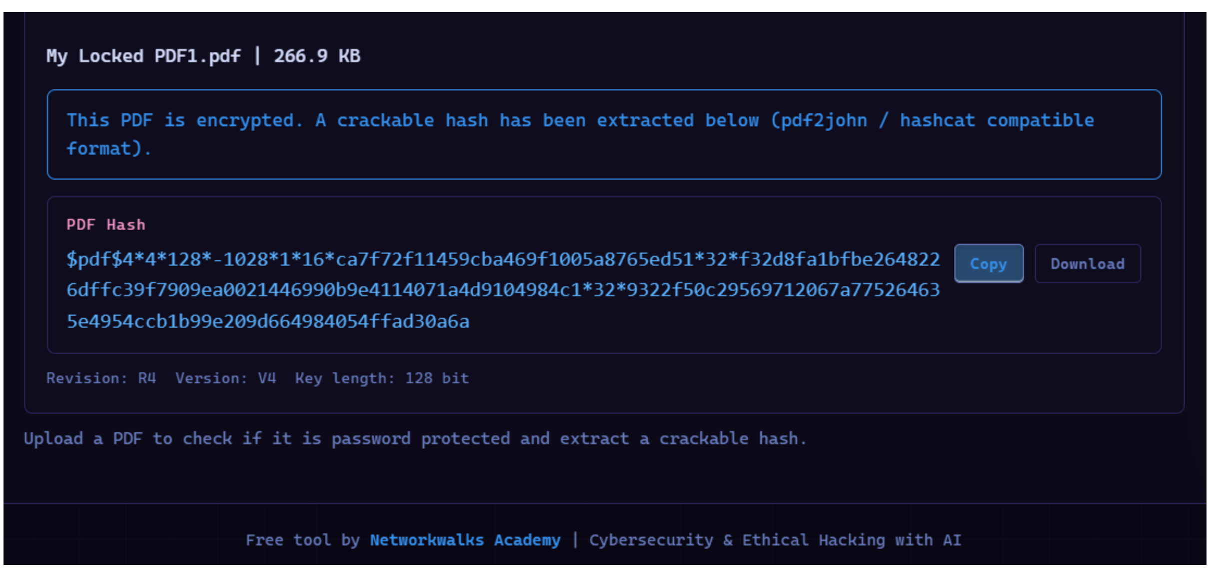
STEP 3

I Upload the locked PDF files to the Hash Calculator. The tool read the files and gave the hash value which started with pdf...

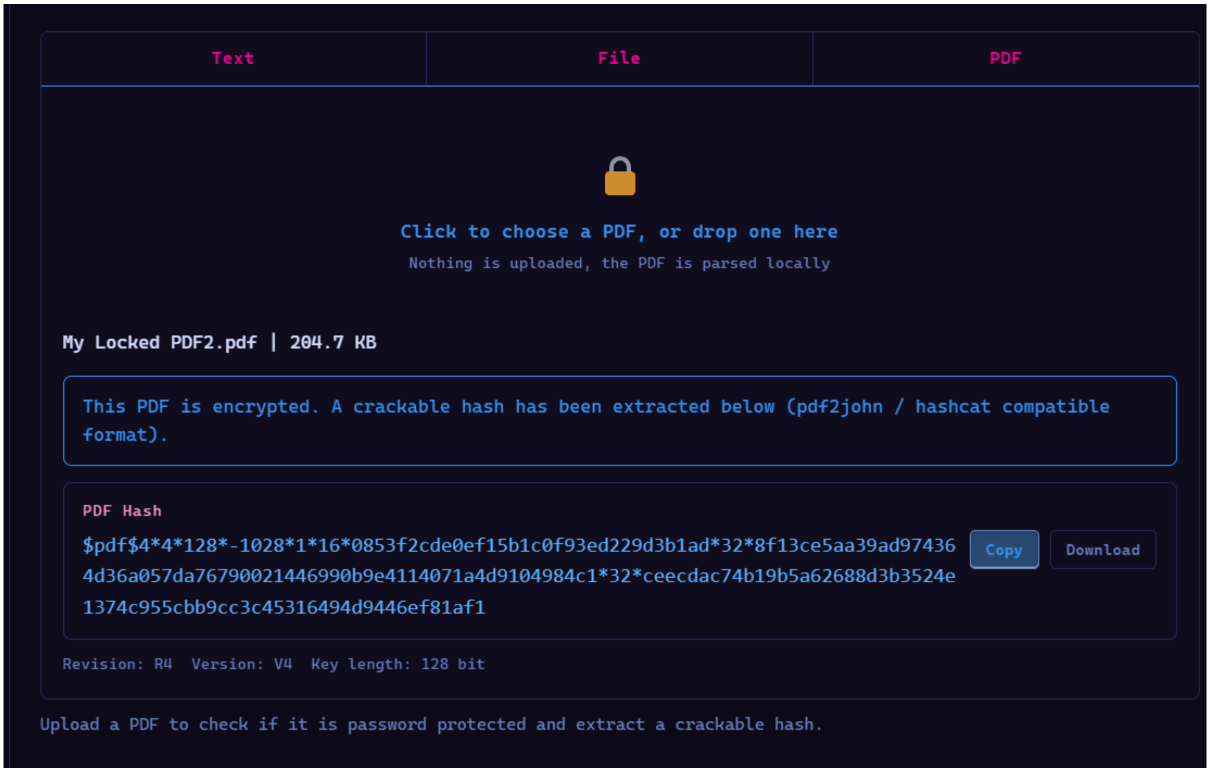


STEP 4

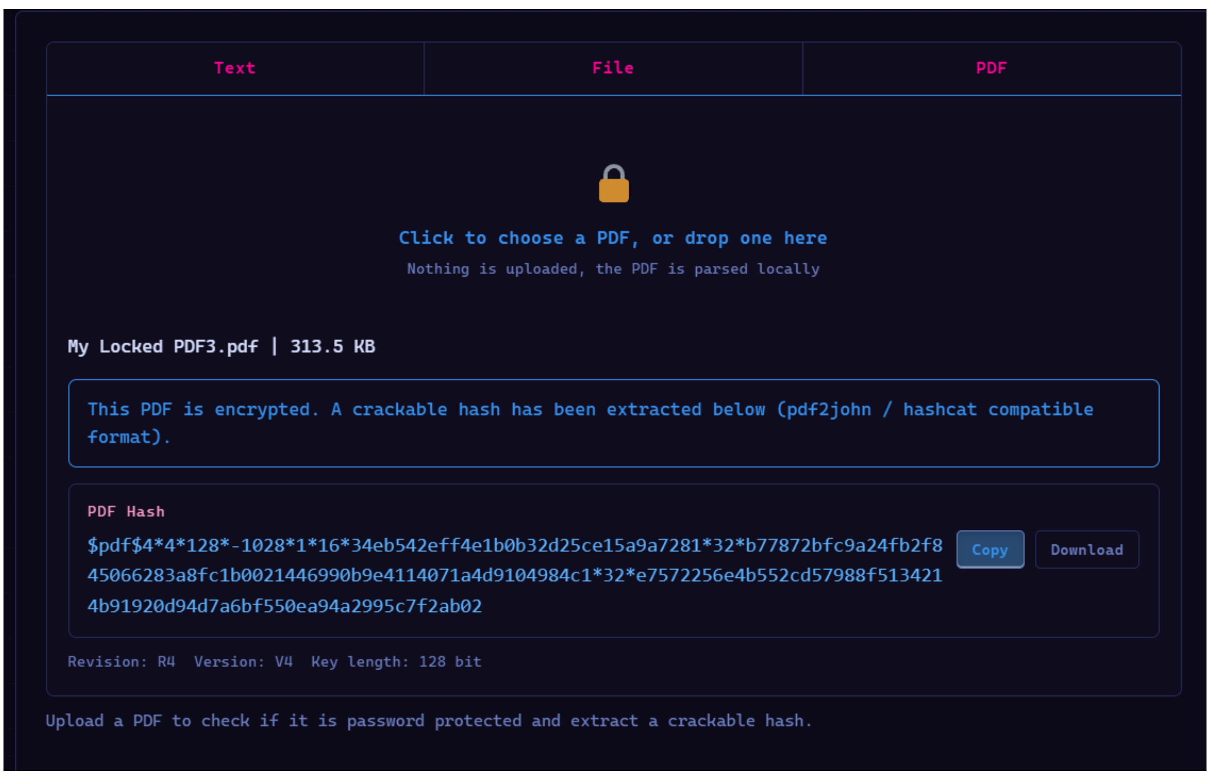
I copied the full hash value of all the locked files as indicated below:



My Locked PDF1 hash value;



My Locked PDF2 hash value;

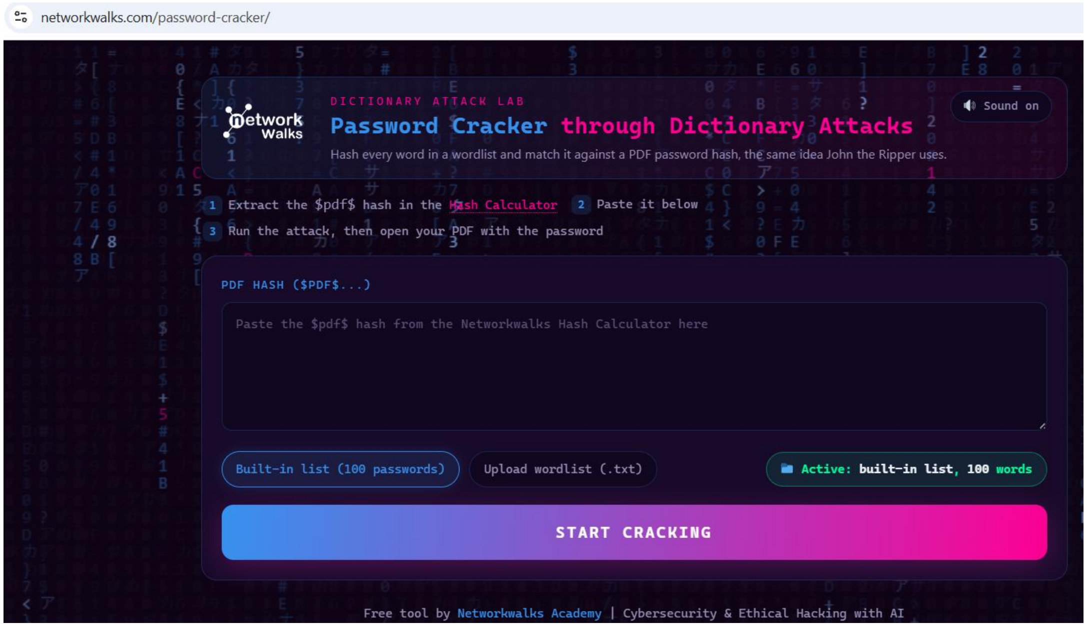


My Locked PDF3 hash value;

STEP 5

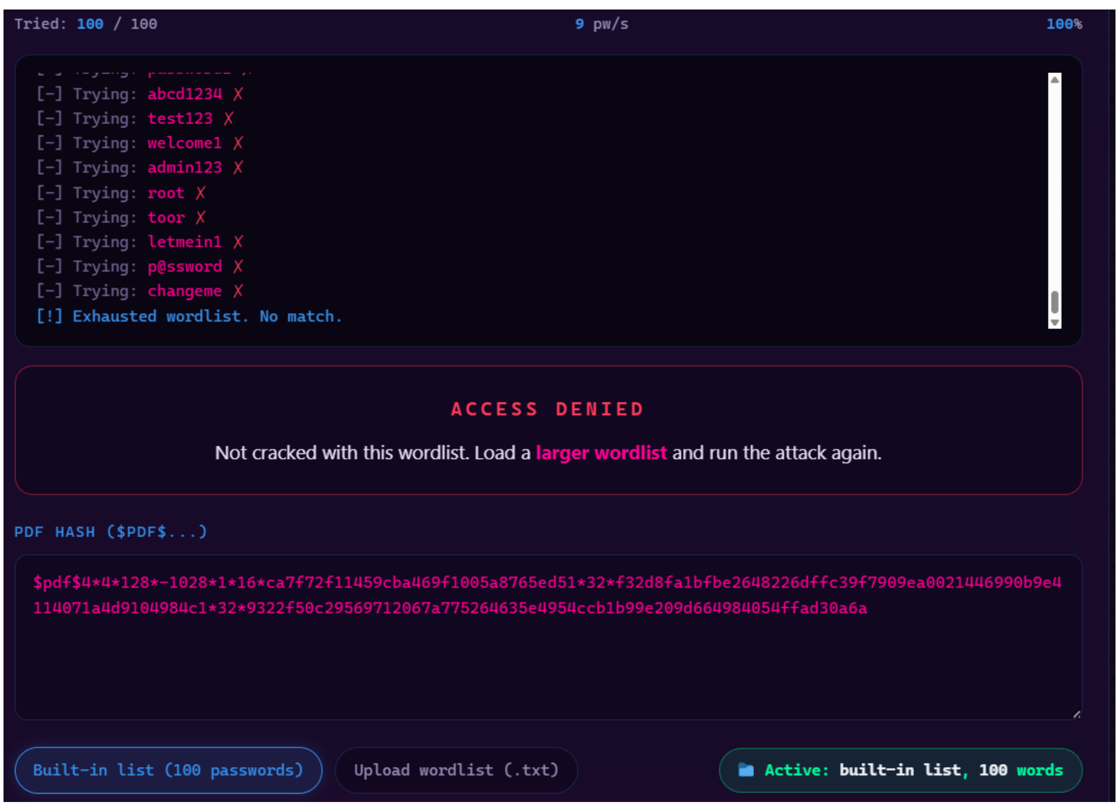
There after i opened the Networkwalks Password Cracker in my web browser using the below link:

https://networkwalks.com/password-cracker/

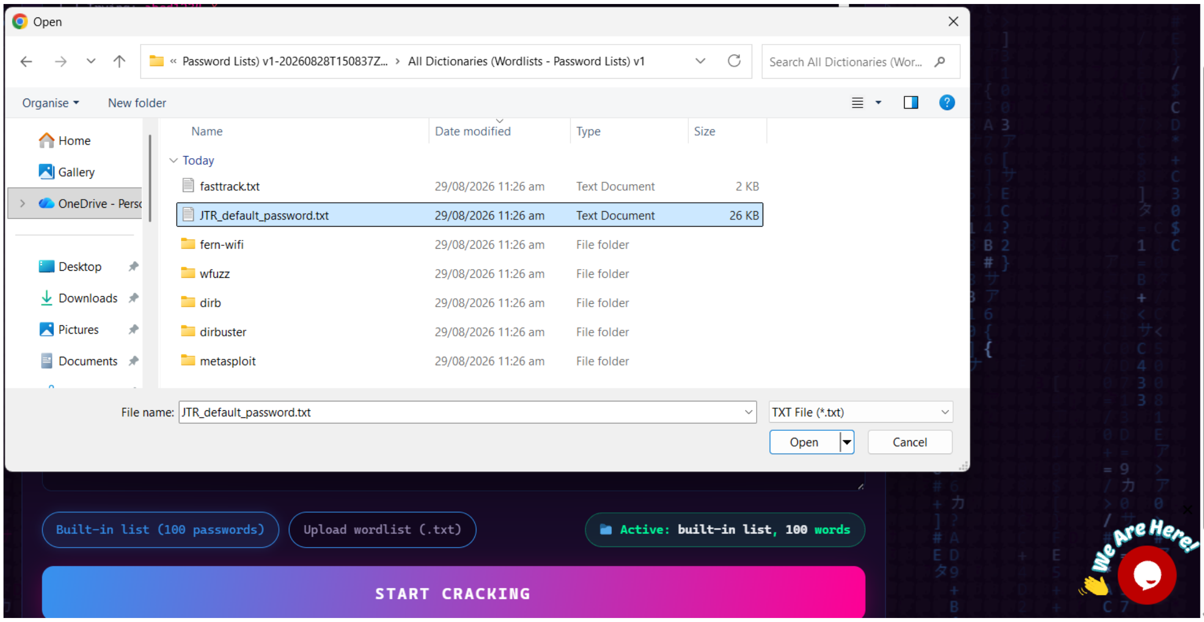


STEP 6

I then pasted the hash value into the Password Cracker and started the attack. The tool tried various passwords until it found the match.



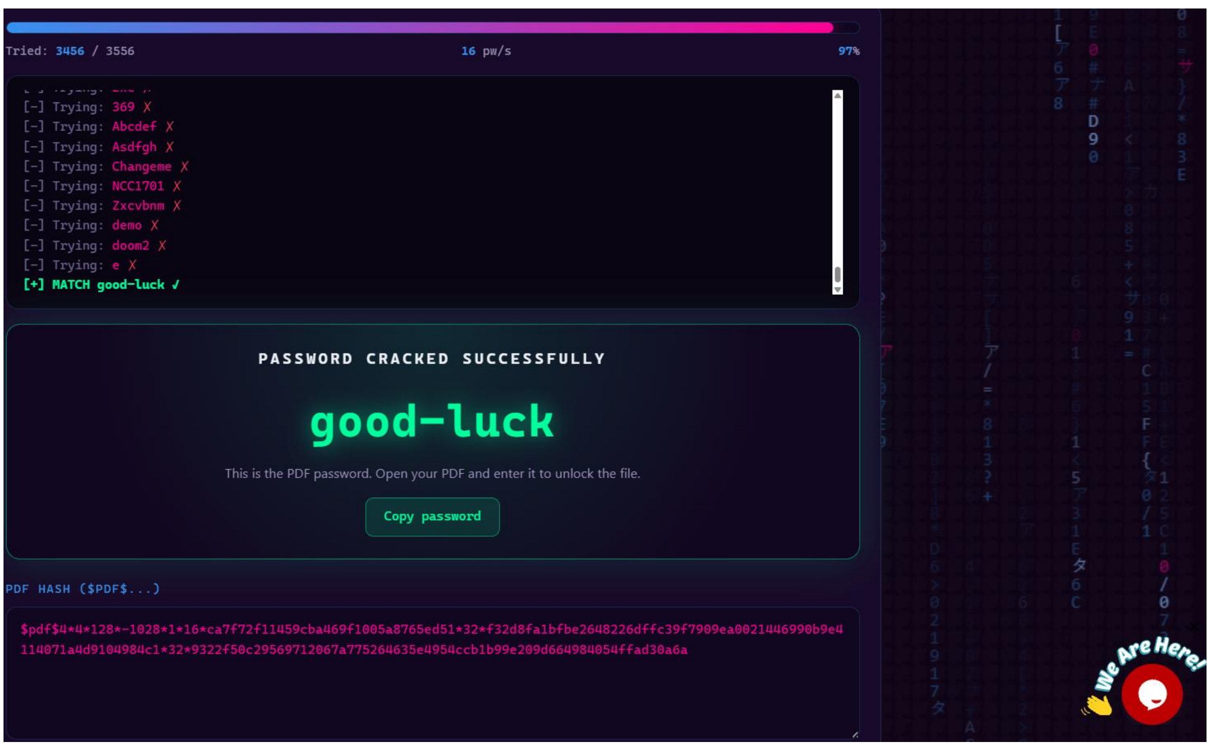
Exhausted the built-in list and no match was found.



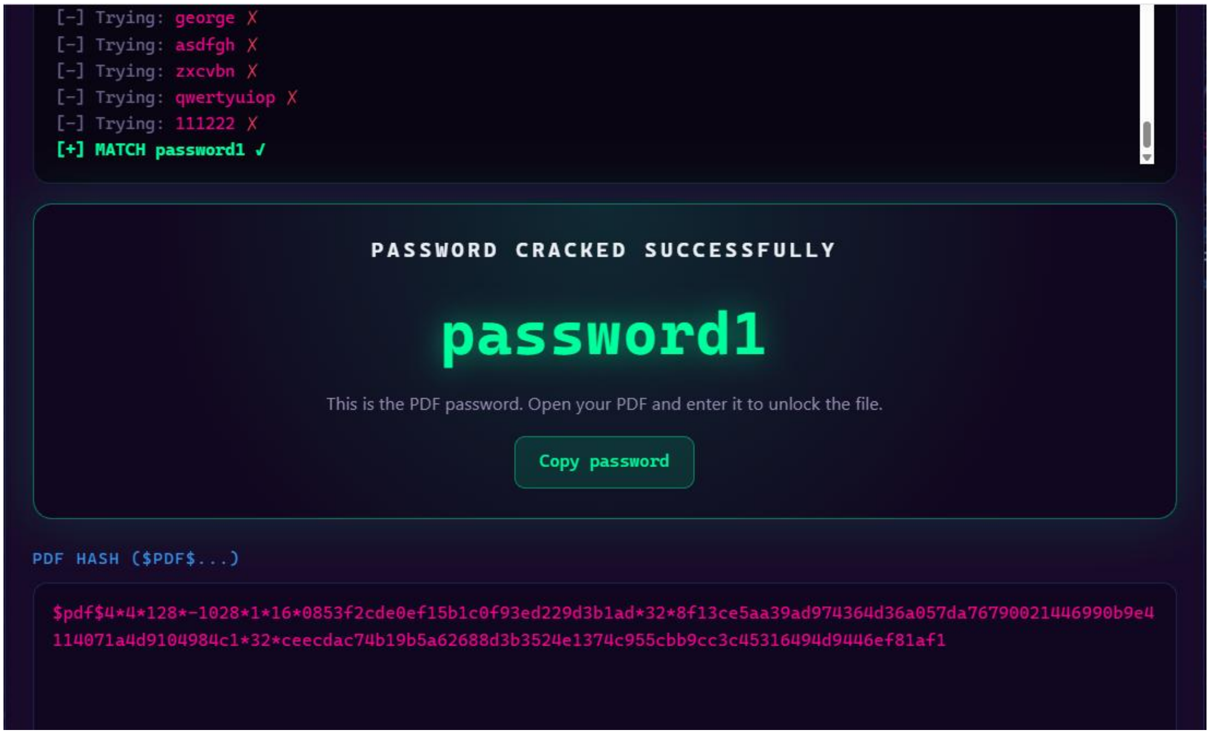
Uploading the JTR default password list.

STEP 7

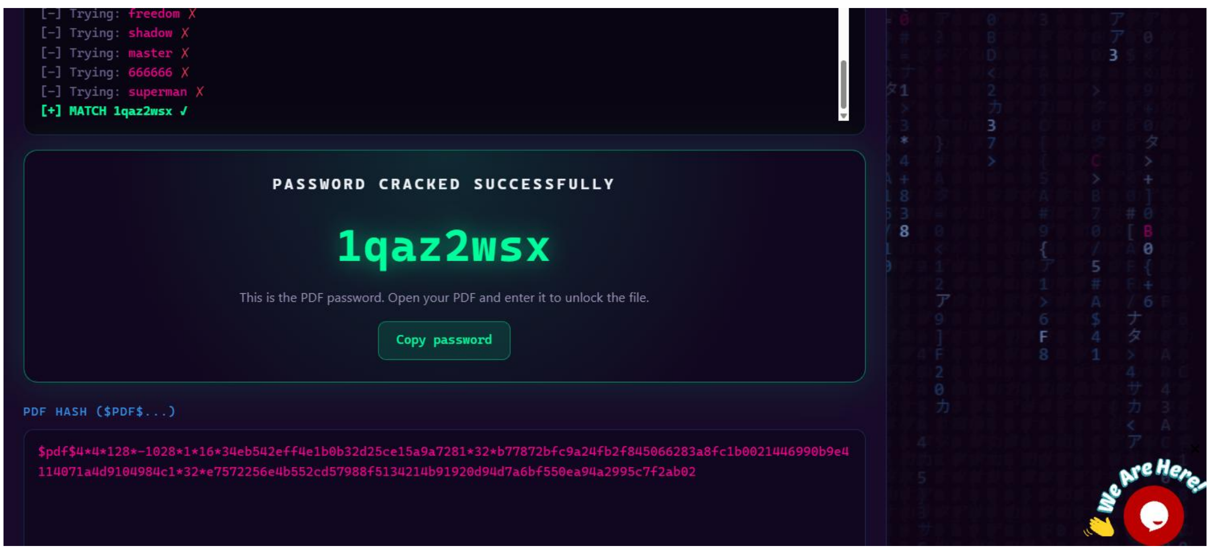
I waited for the tool to finish. The cracked password are indicated in green in the below screenshots.



"My Locked PDF1" password successfully cracked;



"My Locked PDF2" password successfully cracked;



"My Locked PDF3" password successfully cracked;

STEP 8

I open the locked PDF file and enter the cracked password. The results of the opened files were the same as per, 5.1 Module 1: Password cracking with John the Ripper above.

6. Conclusion

This lab helped me understand how password cracking works step by step and why it is important to use strong passwords for protection. I also learned that if a password is short or common, it can be found quickly, which proves the need for strong passwords.

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📌 **Project Information**
Program Name: Cybersecurity program at Networkwalks | **Week:** 03 | **Repository:** GitHub